



Ordinary Differential Equations

Higher-Order Differential Equations

A. Haghighat M.

**Department of Building, Civil and Environmental Engineering
ENGR 213**

Lecture 9 (October 7, 2026)

Today's Lecture

- Theory of Linear Equations
 - Initial-Value and Boundary-Value Problems
 - Homogeneous Equations
 - Nonhomogeneous Equations

Reference

Advanced Engineering Mathematics, D.G. Zill (7th edition), **Sections 3.1**

□ Theory of Linear Equations

❖ Initial-Value Problems

- For a linear differential equation, an n^{th} -order initial-value problem (IVP) is:

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x) \quad (1)$$

subject to: $y(x_0) = y_0, y'(x_0) = y_1, \dots, y^{(n-1)}(x_0) = y_{n-1}$

Theorem – Existence of a unique solution

Let $a_n(x), a_{n-1}(x), \dots, a_1(x), a_0(x)$ and $g(x)$ be continuous on an interval I , and let $a_n(x) \neq 0$ for every x in this interval.

If $x = x_0$ is any point in this interval, then a solution $y(x)$ of the initial-value problem (1) exists on the interval and is unique.

EXAMPLE 1

For the initial-value problem

$$3y''' + 5y'' - y' + 7y = 0 \qquad y(1) = 0, \quad y'(1) = 0, \quad y''(1) = 0$$

verify the existence and uniqueness of the trivial solution $y = 0$.

EXAMPLE 2

For the initial-value problem

$$y'' - 4y = 12x$$

$$y(0) = 4, \quad y'(0) = 1$$

verify that the function $y = 3e^{2x} + e^{-2x} - 3x$ is a solution of the IVP.



❖ Boundary-Value Problems

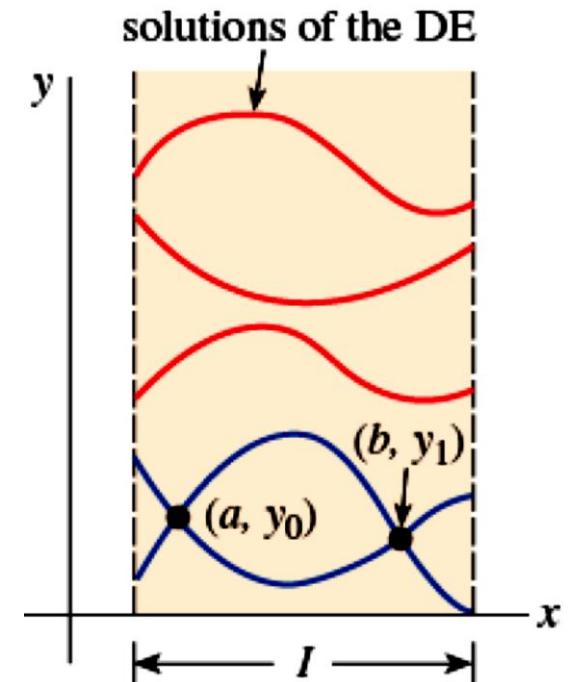
- A linear DE of order two or greater in which the dependent variable y or its derivatives are specified at different points is called a boundary-value problem (BVP).

e.g.,: $a_2(x) \frac{d^2y}{dx^2} + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$ $y(a) = y_0, y(b) = y_1$

- General boundary conditions for a second order BVP:

$$A_1y(a) + B_1y'(a) = C_1$$

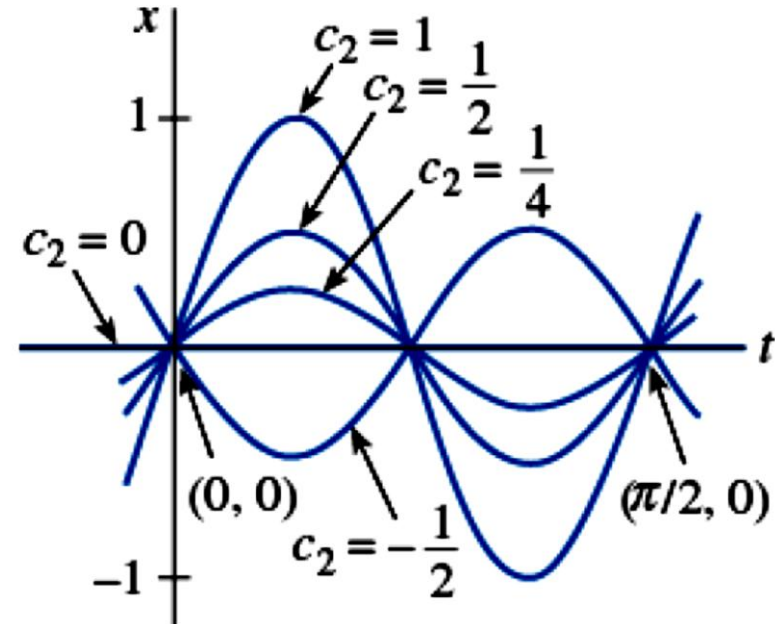
$$A_2y(b) + B_2y'(b) = C_2$$



EXAMPLE 3

For $x'' + 16x = 0$, family of solutions: $x = c_1 \cos 4t + c_2 \sin 4t$

(a) Solution of DE for BCs: $x(0) = 0, x(\pi/2) = 0$



For $x'' + 16x = 0$, family of solutions: $x = c_1 \cos 4t + c_2 \sin 4t$

(b) Solution of DE for BCs: $x(0) = 0$, $x(\pi/8) = 0$

(c) Solution of DE for BCs: $x(0) = 0$, $x(\pi/2) = 1$



❖ Homogeneous Equations

- A linear nth-order differential equation of the form

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = 0$$

is said to be **homogeneous**, whereas an equation

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

with $g(x)$ not identically zero is said to be **nonhomogeneous**.

- Keep in mind the following assumptions when working on the above linear equations. On some common interval I :
 - I. the coefficients $a_i(x)$, $i = 0, 1, 2, \dots, n$, are continuous;
 - II. the right-hand member $g(x)$ is continuous; and
 - III. $a_n(x) \neq 0$ for every x in the interval.

■ Differential Operators

- In calculus, differentiation is often denoted by the capital letter D; that is, $dy/dx = \mathbf{D}y$.
- The symbol D is called a differential operator.
- Higher-order derivatives can be expressed in terms of D as well:

$$\frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d^2 y}{dx^2} = D(Dy) = D^2 y \quad \text{and in general} \quad \frac{d^n y}{dx^n} = D^n y$$

- In general, we define an nth-order differential operator to be:

$$L = a_n(x)D^n + a_{n-1}(x)D^{n-1} + \cdots + a_1(x)D + a_0(x)$$

Some properties

$$D(cf(x)) = cDf(x), \text{ } c \text{ a constant}$$

$$D\{f(x) + g(x)\} = Df(x) + Dg(x)$$

$$L\{\alpha f(x) + \beta g(x)\} = \alpha L(f(x)) + \beta L(g(x)), \text{ } \alpha \text{ and } \beta \text{ constants}$$

■ Differential Equations

- Any linear differential equation can be expressed in terms of the D notation.

e.g.,: $y'' + 5y' + 6y = 5x - 3$

- The general linear nth-order DE can be written compactly as:

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = 0$$

$$L(y) = 0$$

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

$$L(y) = g(x)$$

▪ Superposition Principle

- The sum, or **superposition**, of two or more solutions of a homogeneous linear differential equation is also a solution of the DE.

Theorem – Superposition Principle—Homogeneous Equations

Let $y_1, y_2, \dots, y_k$ be solutions of the homogeneous n th-order differential equation on an interval I .

Then the linear combination $y = c_1y_1(x) + c_2y_2(x) + \dots + c_ky_k(x)$, where the c_i , $i = 1, 2, \dots, k$ are arbitrary constants, is also a solution on the interval.

Corollaries to Theorem

(a) A constant multiple $y = c_1y_1(x)$ of a solution $y_1(x)$ of a homogeneous linear DE is also a solution.

(b) A homogeneous linear DE always possesses the trivial solution $y = 0$.



▪ Linear Dependence and Linear Independence

- A set of functions $f_1(x), f_2(x), \dots, f_n(x)$ is said to be **linearly dependent** on an interval I if there exist constants $c_1, c_2, \dots, c_n$, not all zero, such that

$$c_1 f_1(x) + c_2 f_2(x) + \dots + c_n f_n(x) = 0$$

for every x in the interval. If the set of functions is not linearly dependent on the interval, it is said to be **linearly independent**.

e.g.,: If two functions are linearly dependent, then one is a constant multiple of the other. (i.e., $f_1(x)/f_2(x)$ is not a constant on an interval on which $f_1(x)$ and $f_2(x)$ are linearly independent.)

$$c_1 f_1(x) + c_2 f_2(x) = 0 \xrightarrow{\text{assuming } c_1 \neq 0} f_1(x) = (-c_2/c_1) f_2(x)$$

- A set of n functions $f_1(x), f_2(x), \dots, f_n(x)$ is **linearly dependent** on an interval I if at least one of the functions can be expressed as a linear combination of the remaining functions.

e.g.,: Three functions $f_1(x), f_2(x)$, and $f_3(x)$ are linearly dependent on I if at least one of these functions is a linear combination of the other two for all x in I :

$$f_3(x) = c_1 f_1(x) + c_2 f_2(x)$$

▪ Solutions of Differential Equations

- Suppose each of the functions $f_1(x), f_2(x), \dots, f_n(x)$ possesses at least $n-1$ derivatives. The determinant

$$W(f_1, \dots, f_n) = \begin{vmatrix} f_1 & f_2 & \cdots & f_n \\ f_1' & f_2' & \cdots & f_n' \\ \vdots & \vdots & \ddots & \vdots \\ f_1^{(n-1)} & f_2^{(n-1)} & \cdots & f_n^{(n-1)} \end{vmatrix}$$

where the primes denote derivatives, is called the **Wronskian** of the functions.

Theorem – Criterion for Linearly Independent Solutions

Let $y_1, y_2, \dots, y_n$ be n solutions of the homogeneous n th-order DE on an interval I . Then the set of solutions is linearly independent on I if and only if $W(y_1, y_2, \dots, y_n) \neq 0$ for every x in the interval.

EXAMPLE 4

For $x^2y'' + 7xy' + 13y = 0$ $y_1(x) = \frac{\cos(2 \ln x)}{x^3}, y_2(x) = \frac{\sin(2 \ln x)}{x^3}$ on $(0, \infty)$



■ Fundamental Set of Solutions

- Any set $y_1, y_2, \dots, y_n$ of n linearly independent solutions of the homogeneous linear n th-order differential equation on an interval I is said to be a **fundamental set of solutions** on the interval.

Theorem – Existence of a Fundamental Set

There exists a fundamental set of solutions for the homogeneous linear n th-order differential equation on an interval I .

Theorem – General Solution—Homogeneous Equations

Let $y_1, y_2, \dots, y_n$ be a fundamental set of solutions of homogeneous linear n th-order differential equation on an interval I . Then the **general solution** of the equation on the interval is

$$y = c_1 y_1(x) + c_2 y_2(x) + \dots + c_k y_k(x),$$

where $c_i, i = 1, 2, \dots, k$ are arbitrary constants

EXAMPLE 5

For $y'' - 9y = 0$ $y_1(x) = e^{3x}, y_2(x) = e^{-3x}$ on $(-\infty, \infty)$

EXAMPLE 6

For $y''' - 6y'' + 11y' - 6y = 0$ $y_1 = e^x, y_2 = e^{2x}, y_3 = e^{3x}$ on $(-\infty, \infty)$



❖ Nonhomogeneous Equations

- Any function y_p free of equation. For example, it is a straightforward task to show that the constant function $y_p = 3$ is a particular solution of the nonhomogeneous equation $y'' + 9y = 27$.

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

Theorem – General Solution—Nonhomogeneous Equations

Let y_p be any particular solution of the nonhomogeneous linear n th-order differential equation on an interval I , and let $y_1, y_2, \dots, y_n$ be a fundamental set of solutions of the associated homogeneous DE on I . Then the general solution of the equation on the interval is

$$y = c_1 y_1(x) + c_2 y_2(x) + \dots + c_n y_n(x) + y_p(x)$$

where $c_i, i = 1, 2, \dots, n$ are arbitrary constants

▪ Complementary Function

- In the general solution of a nonhomogeneous linear equation

$$y = \underbrace{c_1 y_1(x) + c_2 y_2(x) + \dots + c_n y_n(x)}_{y_c(x), \text{ complementary function}} + \underbrace{y_p(x)}_{\text{particular solution}}$$

EXAMPLE 6

For $y''' - 6y'' + 11y' - 6y = 3x$, $y_p = \frac{-11}{12} - \frac{x}{2}$

Theorem – Superposition Principle—Nonhomogeneous Equations

Let $y_{p1}, y_{p2}, \dots, y_{pk}$ be k particular solutions of the nonhomogeneous linear n th-order DE on an interval I corresponding, in turn, to k distinct functions $g_1, g_2, \dots, g_k$. That is, suppose y_{pi} denotes a particular solution of the corresponding differential equation

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \dots + a_1(x)y' + a_0(x)y = g_i(x)$$

where $i = 1, 2, \dots, k$. Then

$$y_p(x) = y_{p1}(x) + y_{p2}(x) + \dots + y_{pk}(x)$$

is a particular solution of

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \dots + a_1(x)y' + a_0(x)y = g_1(x) + g_2(x) + \dots + g_k(x)$$

EXAMPLE 7

$$y'' - 3y' + 4y = -16x^2 + 24x - 8 + 2e^{2x} + 2xe^x - e^x$$